Operations

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A quick recap

- Well-defined function: 1. Existence of the image. 2. Uniqueness of the image.
  - Injection: $x \neq y$ then $f(x) \neq f(y)$.
  - Surjection: Range = Codomain.
  - Bijections are permutations!
- Identity function: A function $1_X : X \to X$ s.t. $1_{X(x)} = x$ for all x in X.
- Equality of functions, composition of functions.

Homework 2 review

1. Q1
2. Q2
3. Q3
4. Q4
5. Q5

Recap examples

Are they well-defined? Injective? Surjective?

1. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$
2. $f : \mathbb{Z} \to \mathcal{P}(\mathbb{Z})$, $f(x) = x\mathbb{Z}$.
3. $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = x^x$.
4. $f : \mathbb{Q} \to (\mathbb{Z} \times \mathbb{Z})$, If x can be written as $\frac{p}{q}$, then $f(x) = (p, q)$.
5. $f : \mathbb{Z} \to \mathbb{Z}$, For $x \in \mathbb{Z}$, if y is a multiple of x, then $f(x) = y$.

6. $f : \{O, E\} \to \{O, E\}$ where O is the set of all odd integers and E is the set of all even integers. If there exists $y \in x$ and $y + 5 \in O$, then $f(x) = O$. Otherwise, $f(x) = E$.
7. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = a + x$ for any $a \in \mathbb{R}$
8. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = ax$ for any $a \in \mathbb{R}$
9. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(x, y) = x + y$

Recap examples

cf. PCSA Q5.

Let 1_X be the identity function of the set X.

Let S_1 be the set of all possible functions that can be created by composing 1_X any number of times in any order.

What is the value of $|S_1|$?

Recap examples

cf. PCSA Q5.

Let $f : \mathbb{Z} \to \mathbb{Z}$ be a function such that $f(x) = -x$.

Let D_1 be the set of all possible functions that can be created by composing f any number of times in any order.

What is the value of $|D_1|$?

Recap examples

cf. PCSA Q5.

Let $f : \mathbb{R} \setminus \{0, 1\} \to \mathbb{R} \setminus \{0, 1\}$ be a function such that $f(x) = 1 - x^{-1}$.

Let C_3 be the set of all possible functions that can be created by composing f any number of times in any order.

What is the value of $|C_3|$?

Orders and products

Ordered pairs

Definition 1 (Equality of ordered pairs)

$(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

♣

- $(-3, 5) = (5, -3)$?

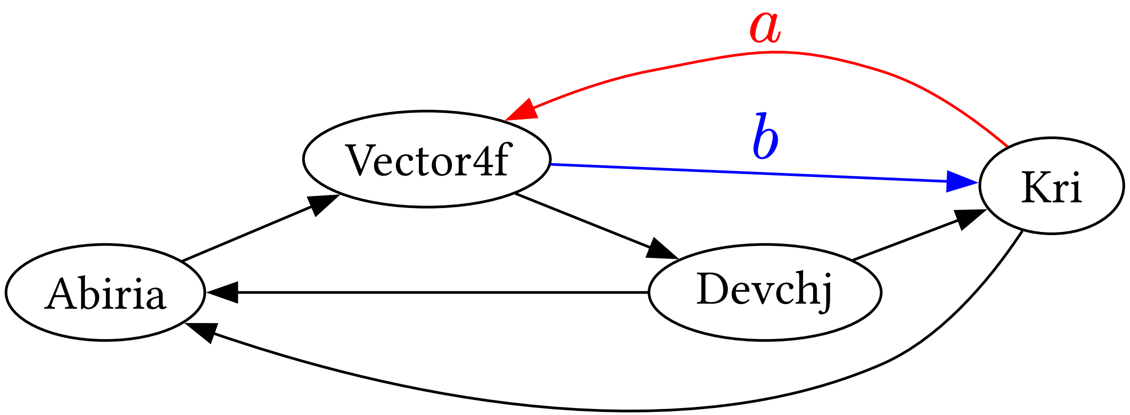
Map each ordered pair to an operation:

- e.g., map (x, y) to $x - y$. Are $(2, 7)$ and $(7, 2)$ mapped to the same result?

Ordered pairs

Example: Directed graphs

- A directed graph is an ordered pair (V, E) where V is the set of vertices and E is the set of edges. Each edge $e \in E$ is again an ordered pair (v_1, v_2) where v_1 and v_2 are vertices.



Ordered pairs

How many ordered pairs do we have?

- cf. PCSA Q6.

Cartesian product

Definition 1 (Cartesian product)

A *Cartesian product* of A and B is the set of all ordered pairs (x, y) where $x \in A$ and $y \in B$. ♣

- cf. PCSA Q7.
- cf. PCSA Q9.

Binary operations are functions!

A set S is given.

Definition 1 (Binary operation)

A function $f : S \times S \to S$ is called a *binary operation* on S.

♣

Note 2

It is often denoted with the infix notation $f(x, y) = x * y$.

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Closedness

As a binary operation is a function, the result of $f(x, y)$ (or $x * y$) should be an element of S if $x, y \in S$.

Examples

1. A standard division is not a binary operation on $\mathbb{Z}$.
   - Counterexample: $5 \div 7 \notin \mathbb{Z}$ even though $5, 7 \in \mathbb{Z}$.
2. A subtraction is not a binary operation on $\mathbb{Z}^+$.
   - Counterexample: $5 - 7 \notin \mathbb{Z}^+$ even though $5, 7 \in \mathbb{Z}^+$.

Examples of binary operations

1. Addition on $\mathbb{Z}$, $x + y$.
2. Subtraction on $\mathbb{Z}$, $x - y$.
3. Multiplication on $\mathbb{Z}$, xy.
4. Exponentiation on $\mathbb{Z}$, x^y.
5. Division on $\mathbb{R}$, $x \div y$.
6. Max on $\mathbb{R}$, $\max(x, y)$.
7. gcd on $\mathbb{Z}^+$, $\gcd(x, y)$.

8. mod on $\mathbb{Z}^+$, $\mathrm{mod}(x, y)$.
9. Average on $\mathbb{Q}$, $(x + y)/2$.
10. Multiplication on $\mathbb{P}$, the set of all primes, pq.
11. Addition on $10\mathbb{Z}$, the set of all multiples of 10, $x + y$.
12. Vector addition on $\mathbb{R}^4$, the 4-dimensional Euclidean vector space, $v + w$.
13. Inner product on $\mathbb{R}^4$, $v \cdot w$.
14. Matrix multiplication on M_4, the set of all 4×4 matrices over $\mathbb{R}$, AB.

15. Addition on $\mathbb{R} \setminus \mathbb{Q}$, the set of all irrational numbers, $x + y$.
16. Intersection on $\mathcal{P}(X)$, the power set of a set X, $A \cap B$.
17. Union on $\mathcal{P}(X)$, $A \cup B$.
18. Cartesian product on $\mathcal{P}(X)$, $A \times B$.
19. Set difference on $\mathcal{P}(X)$, $A \setminus B$.
20. Symmetric difference on $\mathcal{P}(X)$, $A \triangle B$.
21. XOR on $\{0, 1\}^{32}$, the set of all 32-bit binary strings, $x \oplus y$.
22. Concatenation on Σ^*, the set of all strings, $x \mathbin{++} y$.

23. Conditional on $\mathscr{P}$, the set of all false propositions, $p \to q$.
24. Tetration on $\mathbb{Z}^+$, $x \uparrow\uparrow y$.
25. Multiplication on μ_{1212}, the set of all 1212-th roots of unity. xy.
26. Addition on μ_{1212}, $x + y$.
27. Set addition on $\mathcal{P}(\mathbb{Z})$, $A + B$.
28. Function composition on $\mathbb{Z}^{\mathbb{Z}}$, the set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$, $g \circ f$.
29. Function composition on S_2, the set of all bijective functions from $\{0, 1\}$ to $\{0, 1\}$, $g \circ f$.

30. Function composition on $\mathbb{Z}^{\mathbb{Z}}$, the set of all injective functions from $\mathbb{Z}$ to $\mathbb{Z}$, $g \circ f$.

Commutativity

Definition 1 (Commutative binary operation)

A binary operation $f : S \times S \to S$ is said to be *commutative* if $f(x, y) = f(y, x)$ for all $x, y \in S$. ♣

Note 2

In the infix notation, $x * y = y * x$ for all $x, y \in S$. ♢

Notice & Solution time